Coffee, ready-to-drink: HMTc Standard

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CORRESPONDENCE

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KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

This revision 2026-08.

Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for ready-to-drink coffee in Beverages (canned and bottled coffee, cold-brew and nitro coffee, ready-to-drink coffee drinks), evaluated on the liquid as consumed at single strength. Every panel analyte carries a limit: eight numeric ceilings (Hong Kong's direct coffee-beverage lead limit, the drinking-water quality standards read across for the water-borne metals, and the canned-beverage tin limit) plus three species bounded by their parent total. The dry coffee commodity is certified separately in its own as-sold basis; the two bases are never mixed.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit-only): the strictest applicable sovereign maximum level for this exact beverage form and analyte in the row's declared native basis, with disclosed read-across where the exact cell is unregulated, a substantiated not-material determination where a species is bounded or is a method artifact, and a control where no defensible sovereign number and no supportable near-neighbour exist. Never occurrence-derived. Ready-to-drink and dry-commodity bases are never mixed and no brewing, dilution or reconstitution factor is invented.

2 | Master limit table

All values in ppb (µg/kg), as consumed basis. A brand meeting every value qualifies for the mark.

ANALYTE	TIER	HMTC LIMIT	BASIS
Lead (Pb)	Tier 1	200	government maximum level

Cadmium (Cd)	Tier 1	2	human-grade read-across (AU)
Total arsenic (tAs)	Tier 2	10	human-grade read-across (European Union)
Inorganic arsenic (iAs)	Tier 1	≤ tAs	bounded by tAs (reflex speciation)
Mercury, total (tHg)	Tier 1	1	human-grade read-across (European Union)
Methylmercury (MeHg)	Tier 1	≤ tHg	bounded by tHg (reflex speciation)
Nickel (Ni)	Tier 2	20	human-grade read-across (European Union)
Tin (Sn)	Tier 2	100,000	human-grade read-across (European Union)
Aluminium (Al)	Tier 2	200	human-grade read-across (European Union)
Total chromium (Cr)	Tier 2	50	human-grade read-across (AU)
Hexavalent chromium (Cr-VI)	Tier 2	10	human-grade read-across (US-CA)

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	200	government maximum level
Cadmium (Cd)	2	AU — human-food analog
Total arsenic (tAs)	10	European Union — human-food analog

Inorganic arsenic (iAs)	≤ tAs	Inorganic arsenic carries no coffee-beverage-specific sovereign cell; it is a component of total arsenic bounded by the total-arsenic ceiling adjudicated for this row (iAs <= tAs = 10 ug/kg), an enforceable ceiling via reflex speciation.
Mercury, total (tHg)	1	European Union — human-food analog
Methylmercury (MeHg)	≤ tHg	Methylmercury has no formation pathway in a water-based coffee beverage and is a component of total mercury (MeHg <= tHg), bounded by the total-mercury ceiling adjudicated for this row (1 ug/kg); no separate methylmercury ceiling is published.
Nickel (Ni)	20	European Union — human-food analog
Tin (Sn)	100,000	European Union — human-food analog
Aluminium (Al)	200	European Union — human-food analog
Total chromium (Cr)	50	AU — human-food analog
Hexavalent chromium (Cr-VI)	10	US-CA — human-food analog

4 | Cite this standard

Pendergrass, K. (2026). *Coffee, ready-to-drink: HMTc standard*. Institute of Contaminant Standards.

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References

[1] AU — human-food maximum levels
Binds Cd, Cr. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] European Union — human-food maximum levels
Binds tAs, tHg, Ni, Sn, Al. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[3] US-CA — human-food maximum levels
Binds Cr-VI. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

STATUS AND LICENCE

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/coffee-ready-to-drink. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

PUBLICATION HISTORY

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